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0. Setup (repo-relative paths, same convention as hw1_q1.m)

```
clear; clc; close all;
rng(1);
repoDir = fileparts(mfilename('fullpath'));
errDir = fullfile(repoDir, 'ErrP_data_scripts', 'ErrP_data_scripts');
miDir = fullfile(repoDir, 'MI_data_scripts', 'MI_data_scripts'); %
fallback_topoplot_only
outDir = fullfile(repoDir, 'output_q2');
assert(isfolder(errDir), 'ErrP folder not found: %s', errDir);
if ~isfolder(outDir), mkdir(outDir); end
addpath(errDir);
which topoplot
repoCommitRead = 'f0c2085e59ec6c4d9dbb953bc0101a835ad43e10'; % main HEAD
when inputs were read
fprintf('Repo commit read: %s\nMATLAB %s\n', repoCommitRead, version);
```

```
C:\Users\bravo\OneDrive - The University of Texas at
Austin\Desktop\Tanai's Data\UT Undergrad ALL\UT Classes
ALL\BCI\HW_1_export\ErrP_data_scripts\ErrP_data_scripts\topoplot.m
Repo commit read: f0c2085e59ec6c4d9dbb953bc0101a835ad43e10
MATLAB 25.2.0.3055257 (R2025b) Update 2
```

1. Parameters

```
winSec      = [-0.2 0.8];      % analysis window (s) per assignment
baseSec     = [-0.2 0];       % baseline window (s), [start, end]
doBaseline  = true;
ernWin      = [0.05 0.40];    % PROVISIONAL ERN search window (s): confirm on
plotted waveform
peEnd       = 0.80;           % Pe searched from ERN to this time (s)
ernRule     = 'min';          % 'min' = most negative prominent minimum,
'first' = first prominent minimum
promFrac    = 0.10;           % min prominence as a fraction of post-0 peak-
to-peak range
```

```

ccaOnCAR    = false;           % CCA on baseline-corrected non-CAR data by
default
nComp       = 5;
tol         = 1e-9;

```

2. Load and validate

```

D = load(fullfile(errDir, 'ErrP_data_HW1.mat'));
assert(isfield(D, 'trainingEpochs'), 'Expected trainingEpochs');
E = D.trainingEpochs;
X = E.rotation_data;  label = E.label(:);  mag = E.magnitude(:);
C = load(fullfile(errDir, 'ErrP_channels.mat'));
P = C.params;  CL = P.chanlocs;  fs = P.fsamp;  t = P.epochTime(:);
[nT, nCh, nTr] = size(X);
assert(isequal([nT nCh nTr], [1024 32 320]), 'Unexpected size %s', mat2str([nT
nCh nTr]));
assert(nCh == numel(CL), 'Channels (%d) ~= chanlocs (%d)', nCh, numel(CL));
assert(isequal(P.eegChannels(:), 1:nCh), 'Not all 32 channels are EEG');
assert(numel(t)==nT && abs(median(diff(t))-1/fs)<tol, 'Time axis mismatch');
assert(abs(t(P.epochOnset))<tol, 'epochOnset is not t = 0');
assert(~any(isnan(X(:))), 'NaN in data');
assert(numel(label)==nTr && numel(mag)==nTr, 'label/magnitude length
mismatch');
assert(all(ismember(label, [0 1])), 'Unexpected labels');
assert(all(mag(label==0)==0) && all(mag(label==1)>0), 'label/magnitude
inconsistent');
chanNames = strtrim(arrayfun(@(c) char(c.labels), CL, 'UniformOutput', false));
czIdx = find(strcmpi(chanNames, 'Cz'));  fczIdx =
find(strcmpi(chanNames, 'FCz'));
assert(isscalar(czIdx) && isscalar(fczIdx), 'Cz/FCz lookup failed');
fprintf('fs %g Hz | Cz idx %d | FCz idx %d | t(%d)=%.4f s | t(1)=%.4f,
t(end)=%.4f\n', ...
    fs, czIdx, fczIdx, P.epochOnset, t(P.epochOnset), t(1), t(end));
fprintf('Units: not stated in file; |x| median/99%/99.9% =
2.62/12.38/17.83 -> plotting as a.u.\n');
disp('--- params.spectralFilter / spatialFilter / resample (check for prior
CAR or other filtering) ---');
disp(P.spectralFilter); disp(P.spatialFilter); disp(P.resample);

fs 512 Hz | Cz idx 15 | FCz idx 10 | t(512)=0.0000 s | t(1)=-0.9980,
t(end)=1.0000
Units: not stated in file; |x| median/99%/99.9% = 2.62/12.38/17.83 ->
plotting as a.u.
--- params.spectralFilter / spatialFilter / resample (check for prior CAR or
other filtering) ---
    freqs: [1 15]
    order: 2
         a: [1 -3.7538 5.2923 -3.3229 0.7843]
         b: [0.0066 0 -0.0131 0 0.0066]

    type: 'CCA'
    time: [615 616 617 618 619 620 621 622 623 624 625 626 ... ] (1×308
double)

```

```

nComp: 3

is_compute: 1
    ratio: 16
    time: [615 616 617 618 619 620 621 622 623 624 625 ... ] (1×308
double)

```

3. Window and baseline

```

win      = t >= winSec(1)-tol & t <= winSec(2)+tol;
tw       = t(win); nWin = nnz(win); winIdx = find(win);
baseIdx  = t >= baseSec(1)-tol & t < baseSec(2)-tol;
fprintf('Window [%3f, %3f] s: %d samples (first %4f s, last %4f s)
\n',winSec(1),winSec(2),nWin,tw(1),tw(end));
fprintf('Baseline [%2f, %2f] s: %d samples | doBaseline =
%d\n',baseSec(1),baseSec(2),nnz(baseIdx),doBaseline);
Xb = X;
if doBaseline, Xb = X - mean(X(baseIdx,:),:),1); end

Window [-0.200, 0.800] s: 512 samples (first -0.1992 s, last 0.7988 s)
Baseline [-0.20, 0.00] s: 102 samples | doBaseline = 1

```

4. CAR (computed on epochs before averaging)

```

Xc = Xb - mean(Xb,2);
carMaxMean = max(abs(mean(Xc(win,:),:),2)),[],'all');
assert(carMaxMean < 1e-9,'CAR channel mean not ~0');
fprintf('CAR: max |mean across %d channels| = %.2e\n',nCh,carMaxMean);
dsName = {'raw','car'}; dsTitle = {'No spatial filter','CAR'}; dsData =
{Xb,Xc};

CAR: max |mean across 32 channels| = 1.05e-14

```

5. Per-magnitude Cz grand average, ERN/Pe detection, figures

```

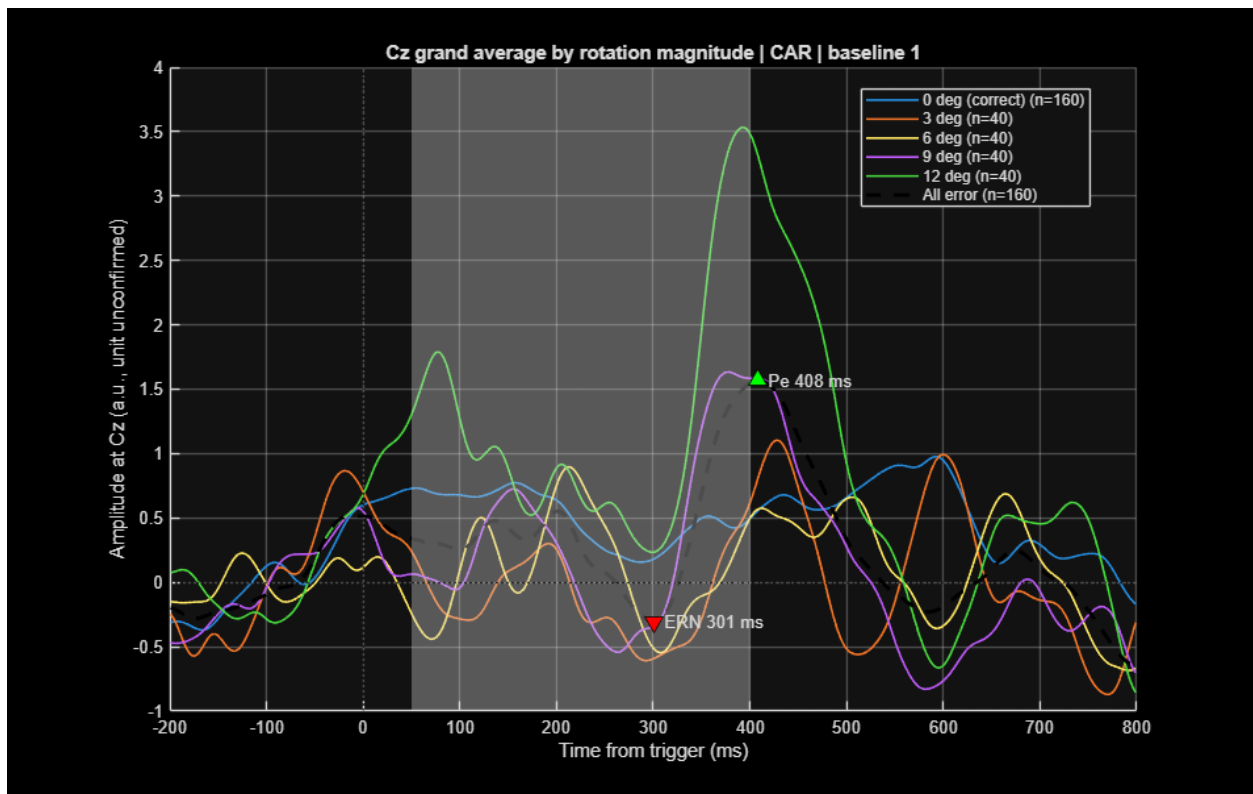
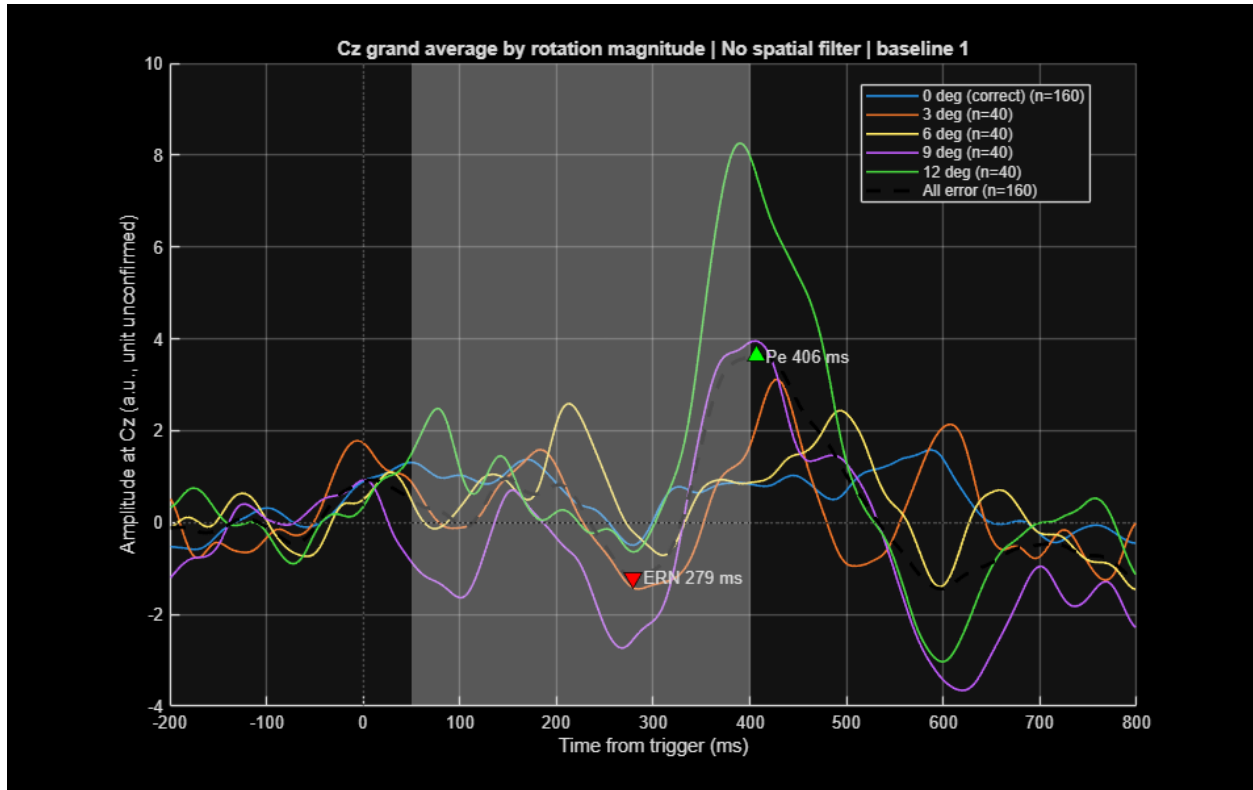
magList = unique(mag)'; errIdx = label==1;
groups = {'All error', errIdx};
for m = magList(magList>0)
    groups(end+1,:) = {sprintf('%d deg error',m), mag==m}; %#ok<SAGROW>
end
rows = {}; DET = struct();
for f = 1:2
    Xf = dsData{f};
    fig = figure('Name',['Cz GA | ' dsTitle{f}], 'Position',[100 100 950
600]); hold on;
    hs = []; leg = {};
    for m = magList

```

```

        idx = mag==m;
        ga = mean(squeeze(Xf(win,czIdx,idx)),2);
        hs(end+1) = plot(tw*1000,ga,'LineWidth',1.2); %#ok<SAGROW>
        if m==0, leg{end+1} = sprintf('0 deg (correct) (n=%d)',nnz(idx));
%#ok<SAGROW>
        else,      leg{end+1} = sprintf('%d deg (n=%d)',m,nnz(idx)); end
%#ok<SAGROW>
    end
    gaErr = mean(squeeze(Xf(win,czIdx,errIdx)),2);
    hs(end+1) = plot(tw*1000,gaErr,'k--','LineWidth',2);
    leg{end+1} = sprintf('All error (n=%d)',nnz(errIdx));
    for g = 1:size(groups,1)
        w = mean(squeeze(Xf(win,czIdx,groups{g,2})),2);
        d = detectErrP(w,tw,ernWin,ernRule,promFrac,peEnd);
        if g==1, DET.(dsName{f}) = d; end
        rows(end+1,:) =
{dsTitle{f},groups{g,1},nnz(groups{g,2}),d.tERN*1000,d.aERN,d.tPe*1000,d.aPe}
; %#ok<SAGROW>
    end
    d = DET.(dsName{f});
    yl = ylim;
    patch([ernWin fliplr(ernWin)]*1000,[yl(1) yl(1) yl(2) yl(2)], [.85 .85
.85], ...
        'FaceAlpha',0.35,'EdgeColor','none','HandleVisibility','off');
    uistack(findobj(gca,'Type','patch'),'bottom');
    if ~isnan(d.tERN),
plot(d.tERN*1000,d.aERN,'kv','MarkerFaceColor','r','MarkerSize',9,'HandleVisi
bility','off'); text(d.tERN*1000,d.aERN,sprintf(' ERN %.0f
ms',d.tERN*1000)); end
    if ~isnan(d.tPe),
plot(d.tPe*1000,d.aPe,'k^','MarkerFaceColor','g','MarkerSize',9,'HandleVisibi
lity','off'); text(d.tPe*1000,d.aPe,sprintf(' Pe %.0f ms',d.tPe*1000));
end
    xline(0,'k:','HandleVisibility','off');
    yline(0,'k:','HandleVisibility','off');
    xlabel('Time from trigger (ms)'); ylabel('Amplitude at Cz (a.u., unit
unconfirmed)');
    title(sprintf('Cz grand average by rotation magnitude | %s | baseline
%d',dsTitle{f},doBaseline));
    legend(hs,leg,'Location','best'); grid on;
    saveFig(fig,outDir,sprintf('q2_cz_ga_%s',dsName{f}));
end
LAT = cell2table(rows,'VariableNames',
{'Filter','Group','nTrials','ERN_ms','ERN_amp','Pe_ms','Pe_amp'});
disp(LAT);
fprintf('ERN shift after CAR: %.2f ms | Pe shift after CAR: %.2f ms (all-
error GA, Cz)\n', ...
    (DET.car.tERN-DET.raw.tERN)*1000,(DET.car.tPe-DET.raw.tPe)*1000);

```



6. Topoplots at ERN and Pe (all error trials), raw and CAR

```
for f = 1:2
    d = DET.(dsName{f});
    assert(~isnan(d.tERN) && ~isnan(d.tPe), 'ERN/Pe not found for %s; adjust
ernWin/promFrac', dsName{f});
    iE = winIdx(d.iERN); iP = winIdx(d.iPe);
    vE = squeeze(mean(dsData{f}(iE, :, errIdx), 3)); vE = vE(:);
    vP = squeeze(mean(dsData{f}(iP, :, errIdx), 3)); vP = vP(:);
    assert(numel(vE)==nCh && numel(vP)==nCh);
    L = max(abs([vE;vP])); % symmetric shared limits
    [-L L]
    fprintf('%s: ERN %.1f ms (spatial mean %.3g), Pe %.1f ms (spatial mean
%.3g), color limits +/-%.3g\n', ...
        dsTitle{f}, d.tERN*1000, mean(vE), d.tPe*1000, mean(vP), L);
    fig = figure('Name', ['Topo | ' dsTitle{f}], 'Position', [100 100 900 420]);
    subplot(1,2,1); safeTopo(vE, CL, [-L L], miDir); cb = colorbar;
ylabel(cb, 'Amplitude (a.u.)');
    title(sprintf('ERN %.0f ms | %s', d.tERN*1000, dsTitle{f}));
    subplot(1,2,2); safeTopo(vP, CL, [-L L], miDir); cb = colorbar;
ylabel(cb, 'Amplitude (a.u.)');
    title(sprintf('Pe %.0f ms | %s', d.tPe*1000, dsTitle{f}));
    sgtitle(sprintf('Grand average, all error trials (n=%d), symmetric
shared scale', nnz(errIdx)));
    saveFig(fig, outDir, sprintf('q2_topo_ern_pe_%s', dsName{f}));
end
```

No spatial filter: ERN 279.3 ms (spatial mean -1.01), Pe 406.2 ms (spatial mean 2.06), color limits +/-4.13

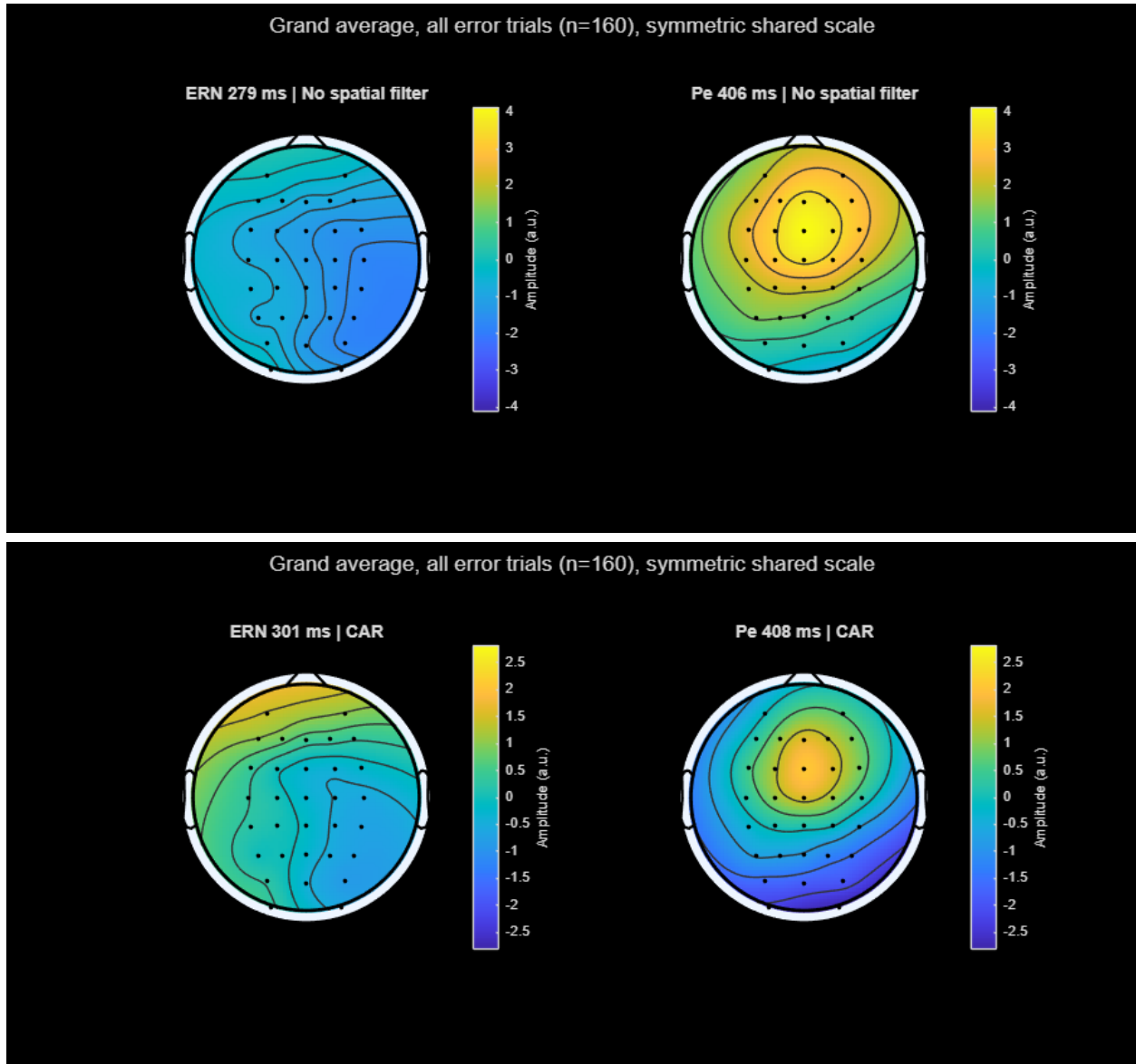
Warning: topoplot failed (Unrecognized function or variable 'icadefs'.); falling back to topoplot_

Warning: topoplot failed (Unrecognized function or variable 'icadefs'.); falling back to topoplot_

CAR: ERN 300.8 ms (spatial mean -9.37e-17), Pe 408.2 ms (spatial mean 2.78e-17), color limits +/-2.83

Warning: topoplot failed (Unrecognized function or variable 'icadefs'.); falling back to topoplot_

Warning: topoplot failed (Unrecognized function or variable 'icadefs'.); falling back to topoplot_



7. CCA spatial filter (error vs correct templates)

NOTE (leakage): descriptive only. Templates and filters are fit on the same trials that are plotted, so canonical correlations are optimistic. No held-out test.

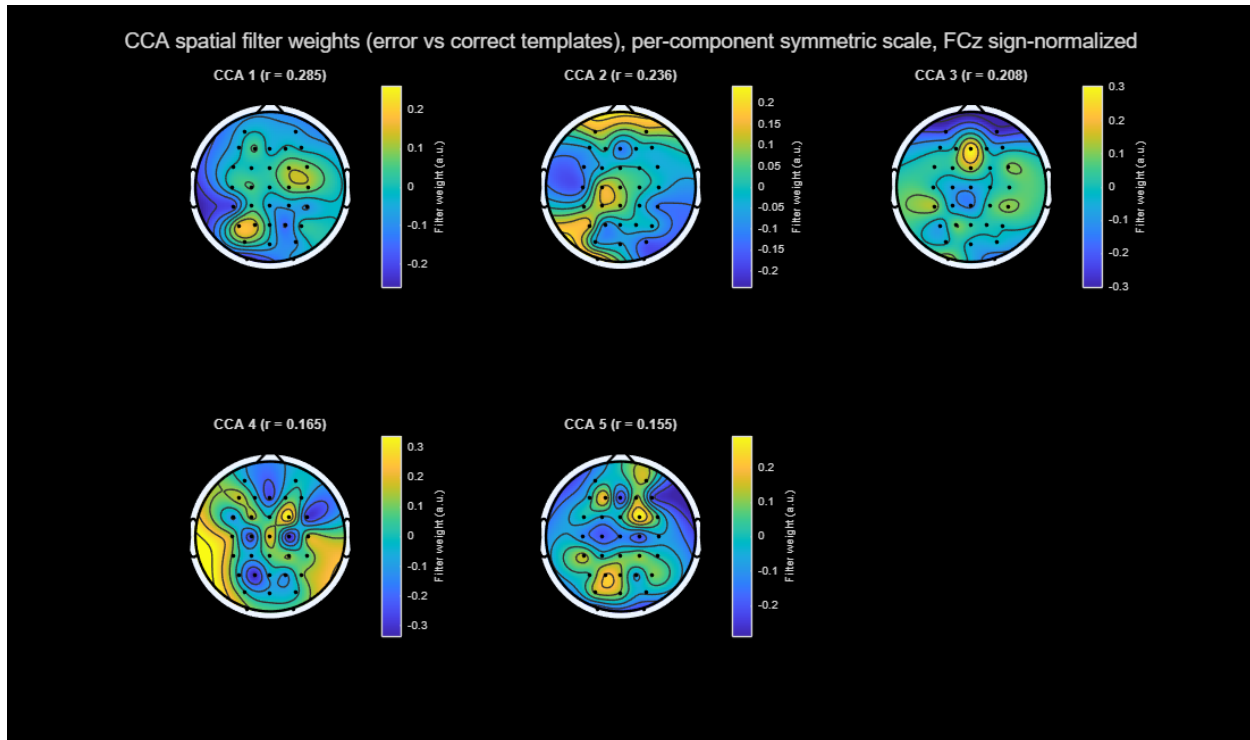
```
Xs = Xb; if ccaOnCAR, Xs = Xc; end
tplErr = mean(Xs(win, :, label==1), 3); % nWin x nCh
tplCor = mean(Xs(win, :, label==0), 3);
concat_data = zeros(nWin*nTr, nCh); concat_ga = zeros(nWin*nTr, nCh);
for k = 1:nTr
    r = (k-1)*nWin + (1:nWin);
    concat_data(r, :) = Xs(win, :, k);
    if label(k)==1, concat_ga(r, :) = tplErr; else, concat_ga(r, :) = tplCor;
end
end
```



```

back to topoplot_
Warning: topoplot failed (Unrecognized function or variable 'icadefs'.);
falling
back to topoplot_
Warning: topoplot failed (Unrecognized function or variable 'icadefs'.);
falling
back to topoplot_

```



8. Save results

```

cnt = array2table([sum(label==0 & mag==0), arrayfun(@(m) sum(label==1 &
mag==m),magList(magList>0))], ...
    'VariableNames',["correct_0deg"
compose("error_%ddeg",magList(magList>0))]);
CCA = table((1:nComp)',rho(:),'VariableNames',{ 'Component','CanonicalCorr'});
R =
struct('commit',repoCommitRead,'matlab',version,'fs',fs,'window_s',winSec,'nW
inSamples',nWin, ...

'doBaseline',doBaseline,'baseSec',baseSec,'ernWin',ernWin,'ernRule',ernRule,'
promFrac',promFrac, ...
    'latencies',LAT,'counts',cnt,'cca_rho',rho,'cca_weights',A,'chanNames',
{chanNames});
save(fullfile(outDir,'results.mat'),'R');
writetable(LAT,fullfile(outDir,'latencies.csv'));
writetable(cnt,fullfile(outDir,'trial_counts.csv'));
writetable(CCA,fullfile(outDir,'cca_corr.csv'));
disp(cnt); disp(CCA);
fprintf('Saved outputs to %s\n',outDir);

```

<i>correct_0deg</i>	<i>error_3deg</i>	<i>error_6deg</i>	<i>error_9deg</i>	<i>error_12deg</i>
160	40	40	40	40
<i>Component</i>	<i>CanonicalCorr</i>			
1	0.28512			
2	0.23567			
3	0.20849			
4	0.16517			
5	0.15534			

Saved outputs to C:\Users\bravo\OneDrive - The University of Texas at Austin\Desktop\tanai's Data\UT Undergrad ALL\UT Classes ALL\BCI\HW_1_export\output_q2

Local functions

```
function d = detectErrP(w,tw,ernWin,ernRule,promFrac,peEnd)
    w = w(:);
    d =
struct('iERN',NaN,'tERN',NaN,'aERN',NaN,'iPe',NaN,'tPe',NaN,'aPe',NaN);
    post = tw > 0;
    prom = promFrac*(max(w(post))-min(w(post)));
    iw = find(tw >= ernWin(1) & tw <= ernWin(2));
    [pk,loc] = findpeaks(-w(iw),'MinPeakProminence',prom);
    if isempty(loc), return; end
    if strcmp(ernRule,'first'), k = 1; else, [~,k] = max(pk); end
    iN = iw(loc(k));
    ia = find(tw > tw(iN) & tw <= peEnd);
    if numel(ia) >= 3
        [~,loc2] = findpeaks(w(ia),'MinPeakProminence',prom);
    else, loc2 = []; end
    d.iERN = iN; d.tERN = tw(iN); d.aERN = w(iN);
    if ~isempty(loc2)
        iP = ia(loc2(1)); d.iPe = iP; d.tPe = tw(iP); d.aPe = w(iP);
    end
end

function safeTopo(v,CL,lim,miDir)
    try
        topoplot(v,CL,'maplimits',lim);
    catch ME
        warning('topoplot failed (%s); falling back to
topoplot_',ME.message);
        addpath(miDir); topoplot_(v,CL,'maplimits',lim);
    end
end

function saveFig(fig,outDir,name)
```

```

exportgraphics(fig,fullfile(outDir,[name '.png']),'Resolution',300);
end

```

<i>ERN_amp</i>	<i>Filter</i> <i>Pe_ms</i>	<i>Pe_amp</i>	<i>Group</i>	<i>nTrials</i>	<i>ERN_ms</i>
	{'No spatial filter'}		{'All error' }	160	279.3
-1.2049	406.25	3.6238			
	{'No spatial filter'}		{'3 deg error' }	40	285.16
-1.4482	427.73	3.115			
	{'No spatial filter'}		{'6 deg error' }	40	310.55
-0.71404	492.19	2.4346			
	{'No spatial filter'}		{'9 deg error' }	40	267.58
-2.7336	404.3	3.9506			
	{'No spatial filter'}		{'12 deg error'}	40	279.3
-0.6412	390.62	8.2544			
	{'CAR' }		{'All error' }	160	300.78
-0.30474	408.2	1.5685			
	{'CAR' }		{'3 deg error' }	40	292.97
-0.60846	427.73	1.1038			
	{'CAR' }		{'6 deg error' }	40	306.64
-0.54561	412.11	0.57512			
	{'CAR' }		{'9 deg error' }	40	263.67
-0.54245	376.95	1.6338			
	{'CAR' }		{'12 deg error'}	40	298.83
0.2334	392.58	3.5329			

ERN shift after CAR: +21.48 ms | Pe shift after CAR: +1.95 ms (all-error GA, Cz)

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